Trevor Dayal

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Seeking internships and co-ops in 2027 and beyond.

Education

Iowa State University of Science and Technology **Expected Graduation June 2029**

Bachelor of Science in Mechanical Engineering

Experience

Simulation & Optimization Engineer | *Formula SAE - Cyclone Racing* **September 2025 – Present**

- Optimized motor design in Motor-CAD using COBYLA and SLSQP.
- Kriged large motor surrogate datasets to reduce optimization time.
- Implemented an AI tools to rapidly iterate code.
- Developed full-vehicle python simulator to model powertrain design.
- Used Optuna and OpenMDAO to optimize motor design.
- Built battery cell count pareto front to determine optimal battery size.
- Prepared Dyno tests to confirm simulation accuracy.
- Wrote 3 DCRs and PDRs and presented work at 2 project showcases.
- Used GitHub to onboard new members.

Team 4192 Captain | *FIRST FRC Robotics* **January 2023- April 2025**

- Captained team 4192 in 2025, managed goal setting and deadlines on a 30-person team to produce a competitive robot in a 6-week period.
- Received the highest award in Texas at the 2025 district championship.
- Led robot design using Onshape to design a robot over a 3-week period.
- Applied lean-manufacturing techniques to reduce fabrication time by >40%.
- Reduced robot cost by over 50% through standardized part design.
- Implemented composite and 3D printed parts to reduce weight and cost.
- Manufactured parts using CNC Router, Waterjet, 3D Printers, and Laser.
- Awarded the highest possible award at 4 events.
- Nominated for FIRST Leadership Award at the 2024 Plano District Event.

Customer Service Representative | *Emler Swim School of Flower Mound* **May 2024- September 2024**

- Engaged customers to provide best experience for their children.

Software

- **CAD:** SolidWorks Modelling, Onshape, NX Designcenter, Ansys Motor-CAD, Inventor, AutoCAD
- **Programming:** Python , Java, Github
- **Fabrication:** Fusion 360 CAM, CNC Router, CO_2 Laser Cutter